

Chain Rule 01: Composite Trigonometric Differentiation

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \sin 5x$ নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — COMPOSITE FUNCTION & CHAIN RULE FORMULATION

The problem represents a composite function of the form $y = f(g(x))$:

$$f(u) = \sin u, \quad u = g(x) = 5x$$

By Leibniz's Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du}(\sin u) \cdot \frac{d}{dx}(5x) = (\cos u) \cdot 5 = 5 \cos 5x$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত সূত্র (Standard Formula)
বহিঃস্থ ফাংশন (Outer)	$\sin u$	$\frac{d}{du}(\sin u) = \cos u$
অভ্যন্তরীণ ফাংশন (Inner)	$u = 5x$	$\frac{du}{dx} = \frac{d}{dx}(5x) = 5$
প্রমিত সাধারণ সূত্র	$\sin(ax)$	$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. **Forgetting Inner Derivative:** The most frequent blunder is writing simply $\cos 5x$ and forgetting to multiply by the inner derivative $\frac{d}{dx}(5x) = 5$.

2. **Angle vs. Coefficient:** $5 \cos 5x \neq \cos(25x)$. The scalar multiplier 5 scales the entire trigonometric amplitude, not the angle.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx}(\sin 5x)$$

ধাপ ২: শৃঙ্খল নিয়ম (Chain Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \cos 5x \cdot \frac{d}{dx}(5x)$$

ধাপ ৩: অভ্যন্তরীণ রৈখিক পদ $5x$ -এর অন্তরজ ৫ বসিয়ে:

$$\frac{dy}{dx} = \cos 5x \cdot 5 = 5 \cos 5x$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = 5 \cos 5x$$

📖 ENGLISH NOTE — ENGINEERING DYNAMICS & PHYSICAL INTERPRETATION

- **Harmonic Oscillation & Velocity:** If $y(t) = \sin(5t)$ describes the displacement of a vibrating spring or alternating AC voltage, its time derivative $v(t) = \frac{dy}{dt} = 5 \cos(5t)$ represents instantaneous velocity.
- **Angular Frequency Scaling:** The angular frequency $\omega = 5 \text{ rad/s}$ emerges as an amplitude multiplier due to the Chain Rule.
- **Domain:** Defined for all real numbers: $\text{Dom}(y) = \text{Dom}\left(\frac{dy}{dx}\right) = \mathbb{R} = (-\infty, \infty)$.

💡 ENGLISH NOTE — FORMAL SUBSTITUTION MODEL

Let $u = 5x \implies \frac{du}{dx} = 5$.

$$y = \sin u \implies \frac{dy}{du} = \cos u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (\cos u) \cdot 5 = 5 \cos 5x$$

This formal substitution is mathematically rigorous and ensures full credit in AMIE exams.

🎯 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Chain Rule Formulation	Writing $\cos 5x \cdot \frac{d}{dx}(5x)$	2.0
Inner Differentiation	Accurate evaluation of $\frac{d}{dx}(5x) = 5$	1.5
Final Standard Format	Standard presentation $5 \cos 5x$	1.5
Total AMIE Section-A Marks		5.0 / 5.0